Exam Date & Time: 26-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

V Sem End Semester Examination Nov/Dec 2024

BIOMATERIALS [BME 3123]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions. Section Duration: 180 mins

1A) An applied strain of 0.4 produces an immediate stress of 10MPa in a piece of rubber, but after 42 days the stress is only 5 MPa. Calculate the relaxation time (τ).

Differentiate creep and relaxation. (4)

1B) A sample of methyl methacrylate [CH_2=C(CH_3)-$COOCH_3$] is polymerized; the resulting polymer has a DP of 1000. Calculate polymer molecular weight. (4)

1C) Why are the tissues in bio prosthetic heart valves treated with glutaraldehyde prior to their clinical use? (2)

2A) Two patients of ages 40 years and 65 years are advised Total hip arthroplasty (THA). Analyze critically, the type of fixation (cemented /cement less) would be appropriate for them. (3)

2B) Differentiate total hip arthroplasty and hemi hip arthroplasty. (2)

2C) Compare and analyse the steps involved for the fixation of total hip replacement and total knee replacement. (5)

3A) Justify with reason how a bi-leaflet valve is superior to caged and ball valve. Explain different types of heart valves with its constructional features and the materials used for construction. (5)

3B) Highlight the limitation of rigid contact lenses (PMMA polymer). Propose how these limitations are rectified using elastomeric lenses (PHEMA polymer). (2)

3C) Silicone rubber lenses despite their good oxygen permeability are not being accepted widely as contact lenses. Analyse the problems associated for such poor acceptance. (3)

4A) Classify dental implants based on design. (3)

4B)	Analyse the two step processes involved for the fixation of dental implant. Justify the need of assessing bone height, bone density prior to implantation.	(5)
4C)	Eraina has uncoated and hydroxyapatite coated Co-Cr based dental implants. Which one would she prefer to use for improving bone cell density?	(2)
5A)	Sameer wishes to make homogeneous porous scaffolds. Why would 3D bioprinting be appropriate for his experiment over porogen leaching and freeze drying? Justify.	(2)
5B)	Why is the telopeptide region of the collagen fibril appeared weak? Susil intends to extract collagen for the fabrication of a composite matrix. What type of isolation technique would be appropriate? Describe the steps in brief and justify the choice. Will the orientation of the collagen fibres make any impact in the design of the composite matrix?	(5)
5C)	Shalini intends to make chitosan based porous scaffold. In this context, Shalini has made chitosan hydrogel. Propose a strategy to assist Shalini to fabricate porous chitosan scaffold (Briefly, elaborate the process involved).	(3)

-----End-----